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A COMPOSITIONAL AND DIAGRAMMATIC SYNTAX FOR PARAMETRIC
LINEAR PROGRAMMING

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Dissertação de Mestrado apresentada ao Programa de Pós-Graduação em Informática, Instituto de Computação e Instituto Tércio Pacitti de Aplicações e Pesquisas Computacionais, Universidade Federal do Rio de Janeiro, como requisito parcial à obtenção do título de Mestre em Informática.

Orientador: Prof. João Paixão, D.Sc.

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RESUMO

Esta dissertação investiga como problemas de otimização podem ser compreendidos dentro de um arcabouço composicional e diagramático unificado, fundamentado na teoria das categorias. A motivação central é a observação de que estruturas matemáticas essenciais em otimização — relações, funções convexas, formas quadráticas, poliedros e programas lineares — podem ser interpretadas como morfismos em categorias nas quais a composição corresponde à eliminação ou minimização de variáveis internas. A partir desta perspectiva, a otimização emerge não como uma coleção de técnicas isoladas, mas como uma operação algébrica inerente à estrutura relacional. O desenvolvimento segue uma progressão incremental, partindo das relações binárias clássicas enriquecidas por quantais de Lawvere. A cada etapa é introduzido um único gerador novo e um conjunto finito de axiomas. A Álgebra Quadrática Gráfica (GQA) representa problemas de minimização da norma ℓ_2 sobre subespaços afins; a Álgebra Poliedral enriquece o arcabouço com restrições de desigualdade linear; e a Álgebra Gráfica para Programação Linear Paramétrica (GLP) constitui a contribuição central da tese, na qual um programa linear é tratado como um morfismo compositável com outros blocos de otimização. Para cada álgebra são demonstrados resultados de solidez e completude com respeito às categorias semânticas correspondentes. Um resultado central mostra que toda função convexa afim por partes admite representação como função valor de um programa linear, e que essa equivalência se manifesta diagramaticamente como uma forma normal canônica, eliminando no nível sintático a distinção entre um LP e uma função convexa. O cálculo diagramático para GLP desempenha, para a programação linear, o mesmo papel que a Álgebra Linear Gráfica desempenha para matrizes: fornece uma linguagem de formas normais, regras de reescrita e teoremas de completude dentro dos quais argumentos de otimização podem ser conduzidos simbolicamente, antes e independentemente de qualquer computação numérica.

Palavras-chave: teoria das categorias; diagramas de corda; programação linear; otimização convexa; categorias monoidais simétricas; relações ponderadas; quantais; álgebra gráfica; formas normais; completude.

ABSTRACT

This dissertation investigates how optimization problems can be understood within a unified compositional and diagrammatic framework grounded in category theory. The central observation is that key mathematical structures in optimization can be interpreted as morphisms in categories whose composition corresponds to the elimination or minimization of internal variables. From this perspective, optimization emerges not as a collection of isolated techniques but as an algebraic operation intrinsic to relational structure.

The development follows an incremental progression rooted in Lawvere quantales and weighted relations, adding one new generator and a finite set of axioms at each step. We recall the theory of Quantale relations and the Polyhedral Algebra framework as preliminaries steps to develop the novel Graphical Linear Programming (GLP), where morphisms – diagrams – are parametric linear programs. Those composes with other optimization blocks, so that the elimination of internal variables, or, in this case, the optimization itself, corresponds exactly to categorical composition. Soundness and completeness results for the previous algebras are recalled from prior works, and in the last section we also present the completeness theorem for our new language.

The diagrammatic calculus plays the same role for linear programming that Graphical Linear Algebra plays for matrices: it provides a expressible language for normal forms and rewriting rules in which arguments can be carried out symbolically, prior to and independently of numerical computation, making the main properties of the system evident in the syntax itself. The tools and results developed and proved here are this diagrammatic calculus for Linear Programming.

Keywords: category theory; string diagrams; linear programming; convex optimization; symmetric monoidal categories; weighted relations; quantales; graphical algebra; normal forms; completeness.

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1 POLYHEDRAL RELATIONS

We now turn to a combinatorially richer class of objects: polyhedra and polyhedral cones. These are the fundamental geometric structures underlying linear programming, and they admit a categorical treatment entirely analogous to the one developed for linear and affine relations. The key difference is that composition here is governed by Fourier-Motzkin Elimination (FME): projecting a polyhedron onto a subspace is precisely variable elimination, and it is this operation that categorical composition will simulate diagrammatically.

The theory developed here remains within the category of binary relations, since morphisms are still indicator functions of subsets of $\mathbb{R}^n \times \mathbb{R}^m$, encoding feasibility constraints rather than objective functions. In this sense, **GPA** does not yet enter the weighted relational world, and so is not yet able to express more meaningful optimization problems. This is nonetheless a critical step in the development. Axiomatizing linear inequality constraints diagrammatically, **GPA** provides the Symmetric Monoidal Inequality Theory (SMIT) [BDS21] that will serve as the structural foundation for our final contribution, **GLP**, where these constraints are combined with a linear objective to form full parametric linear programs.

In what follows we describe and show the complete theory of polyhedral relations and polyhedral cones. Just as **AffRel** captured affine space as morphisms, the categories **PCRel** and **PolyRel** capture polyhedral cones and polyhedra as morphisms, with relational composition playing the role of variable elimination.

1.1 SEMANTICS: THE PROP **PCREL** OF POLYHEDRAL CONES

We begin by recalling the standard definitions of polyhedral cones and the corresponding semantic category.

Definition 1.1. *A set $C \subseteq \mathbb{R}^n$ is called a polyhedral cone if there exists a matrix $A \in \mathbb{R}^{m \times n}$ such that*

$$C = \{x \in \mathbb{R}^n \mid Ax \geq 0\}.$$

Definition 1.2 (The prop **PCRel** of polyhedral cones). *Let k be an ordered field, in this case the real numbers. The prop **PCRel** _{k} is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n .*
- *A morphism $n \rightarrow m$ is a polyhedral cone $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$*

and $A \in k^{p \times (n+m)}$ such that

$$R = \left\{ (x, y) \in k^n \times k^m \mid A \begin{pmatrix} x \\ y \end{pmatrix} \geq 0 \right\}.$$

- *Composition is relational composition.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the Cartesian product.*

Observe that in comparison with the previous categories of **AffRel**, this category is way more general although it does not completely encode the notion of Affine Spaces. However, it does introduce to us a somewhat unorthodox notion of conical polyhedral relations. That is, the fact that two elements x, y may be related to one another by a relation R such that the set of all pairs xRy form a polyhedral cone. Although this description remains very set-theoretically, it starts to flirt with the idea of x and y being apparted from a specific class of maps. More precisely, instead of thinking of the whole geometry of polyhedral cones, one may think of the pairs (x, y) such that $Ax \geq By$ for some A and B fully defined by the relation R .

1.2 SYNTAX: THE ALGEBRA $\mathbf{GPA}_{\perp}$

Definition 1.3 (Graphical Polyhedral Algebra). *We call **GPA** the SMT generated by extending **GAA** with the following generator and its semantics:*

$$\llbracket \text{---} \boxed{\geq} \text{---} \rrbracket = \{(x, y) \mid x, y \in k, x \geq y\}$$

along with the following additional equations:

$$\begin{array}{lll}
 (P0) & \boxed{} & = \text{---} \boxed{\geq} \text{---} \bigcirc \\
 (P1) & \text{---} \boxed{\geq} \text{---} \bullet & \subseteq \text{---} \bullet \begin{array}{l} \swarrow \boxed{\geq} \\ \searrow \boxed{\geq} \end{array} \\
 (P2) & \bigcirc \text{---} \boxed{\geq} \text{---} & = \begin{array}{l} \boxed{\geq} \text{---} \bigcirc \\ \boxed{\geq} \text{---} \bigcirc \end{array} \\
 (P3) & \text{---} \boxed{\geq} \text{---} \bullet & = \text{---} \bullet \\
 (P4) & \bigcirc \text{---} & \subseteq \bigcirc \text{---} \boxed{\geq} \text{---}
 \end{array}$$

$$\begin{aligned}
(P5) \quad k > 0 \quad & \boxed{k} \text{---} \boxed{\geq} \text{---} = \text{---} \boxed{\geq} \text{---} \boxed{k} \\
(P6) \quad k < 0 \quad & \boxed{k} \text{---} \boxed{\geq} \text{---} = \text{---} \boxed{\leq} \text{---} \boxed{k} \\
(antisym) \quad & \begin{array}{c} \text{---} \bullet \text{---} \boxed{\geq} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \boxed{\leq} \text{---} \bullet \text{---} \end{array} \subseteq \text{---} \\
(spider) \quad & \begin{array}{c} \boxed{\geq} \text{---} \bullet \text{---} \boxed{\geq} \\ \boxed{\geq} \text{---} \bullet \text{---} \boxed{\geq} \end{array} = \begin{array}{c} \bullet \text{---} \boxed{\geq} \text{---} \bullet \\ \bullet \text{---} \boxed{\geq} \text{---} \bullet \end{array} \\
(direction) \quad & \boxed{\leq} \text{---} \boxed{\geq} = \text{---} \bullet \quad \bullet \text{---}
\end{aligned}$$

Definition 1.4 (Graphical Polyhedral Cones Algebra). *We call $\mathbf{GPA}_{\vdash}$ the SMT generated by $\mathbf{GPA}$ but excluding the affine generator $\vdash$ and its associated equations. This fragment axiomatises polyhedral cones, where no translation is present.*

Definition 1.5 (The Free Categories of Polyhedral Diagrams). *We call $\text{FreeP}_{\mathbf{GPA}}$ the PROP freely generated by $\mathbf{GPA}$, and $\text{FreeP}_{\mathbf{GPA}_{\vdash}}$ the PROP freely generated by $\mathbf{GPA}_{\vdash}$.*

1.3 THE ORDER STRUCTURE OF THE INEQUALITY GENERATOR

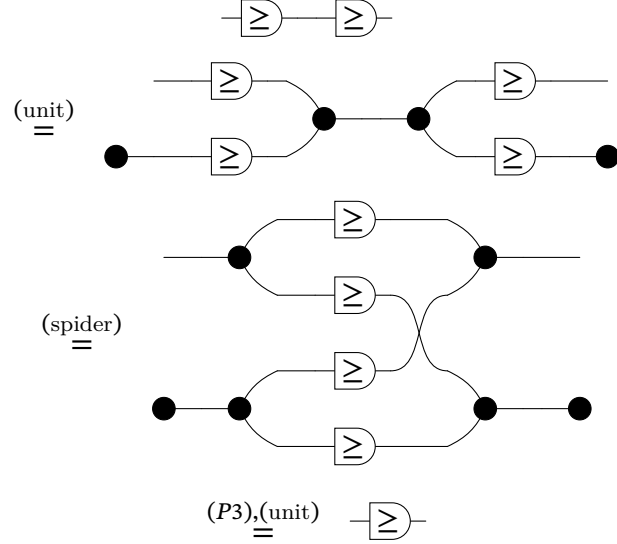
The central computational primitive for reasoning inside this algebra is Fourier–Motzkin Elimination, which projects a polyhedral cone onto a lower-dimensional space by eliminating one variable at a time.

It is worth being explicit about what that algorithm actually does to a system $Ax \geq 0$, because it fixes exactly which diagrammatic rules we have to establish first. To eliminate a variable t , the algorithm divides each row containing t by its coefficient, so that the row becomes a bound on t ; it records whether the bound is an upper or a lower one, according to the sign of that coefficient; and it then replaces the whole collection of bounds by one inequality for each upper–lower pair. Three moves, therefore: rescaling a row, moving a term across a sum, and merging bounds pairwise. Two of them are axioms of Definition 1.3; this section supplies the third and the order laws the axioms omit, and Section 1.4 assembles them into the elimination itself.

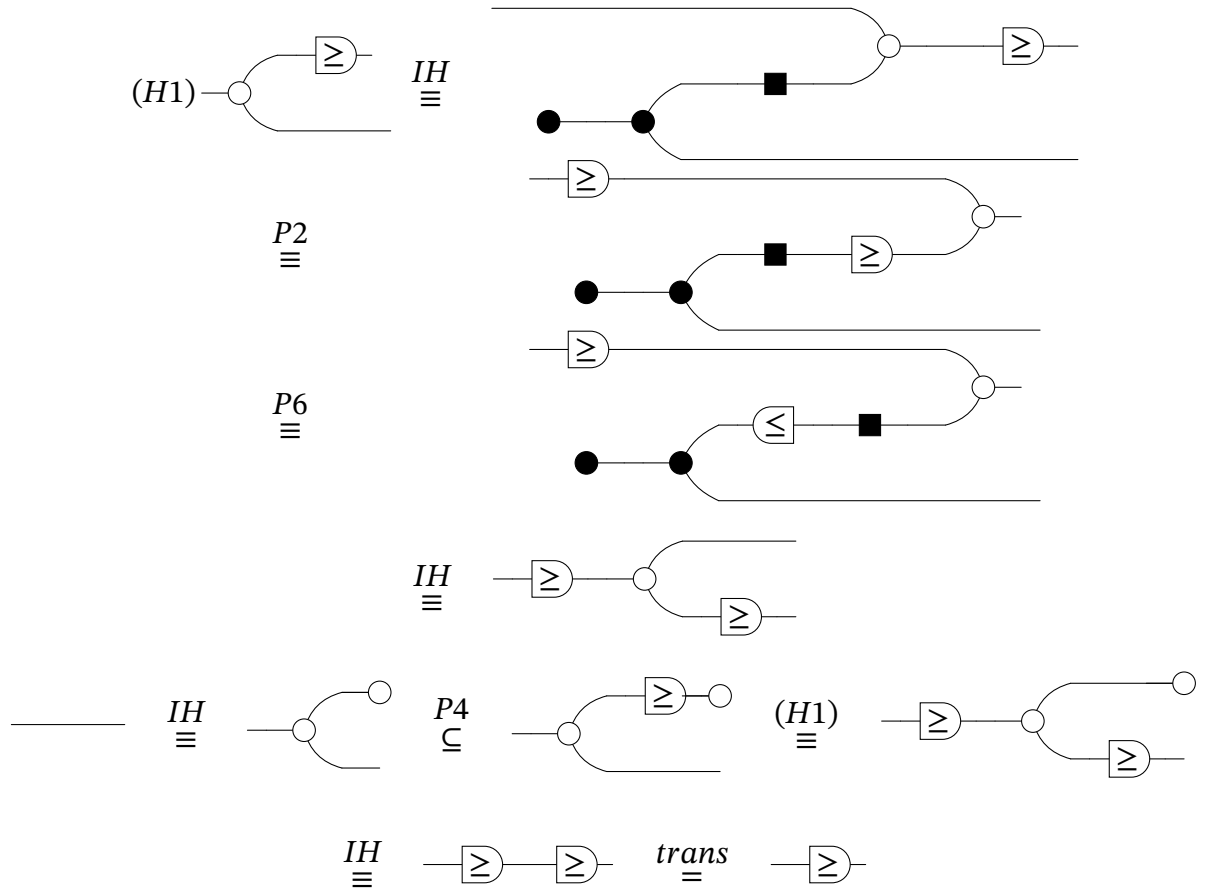
Rescaling and sign are given outright by (P5) and (P6), and the pairwise merge by (spider). What the axioms do not state, and what we need first, is that $\boxed{\geq}$ deserves its name: reflexivity and transitivity are not among them, and antisymmetry is postulated only as the inclusion $\subseteq$.

Lemma 1.1 (The generator is an order). *In $\mathbf{GPA}_{\perp}$ the generator $\dashv\!\!\!\geq$ is reflexive, transitive and antisymmetric.*

Proof. Transitivity is (spider) with one input fed by the black unit and one output discarded, the plugs being absorbed by (P3) and the black (co)unit laws:



For reflexivity we need a previous results that we call *H1*



Antisymmetry, the converse of the inclusion (antisym):

$$\begin{aligned}
(1) \quad & \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\leq} \text{---} \end{array} \bullet \text{---} \subseteq \text{---} \\
(2) \quad & \text{---} = \text{---} \bullet \text{---} \bullet \text{---} \subseteq \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\leq} \text{---} \end{array} \bullet \text{---} \\
(1) \wedge (2) \Rightarrow & \text{---} = \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\leq} \text{---} \end{array} \bullet \text{---}
\end{aligned}$$

□

Since $\text{---} \boxed{\leq} \text{---}$ is $\text{---} \boxed{\geq} \text{---}$ with an antipode on each side, by (P6) at $k = -1$, every axiom comes in a mirrored version, and we use both without further comment.

The move the axioms leave implicit is the transport of a term across a sum: it is the step that turns a row of a system into a bound on a single variable, and it is the content of the next lemma.

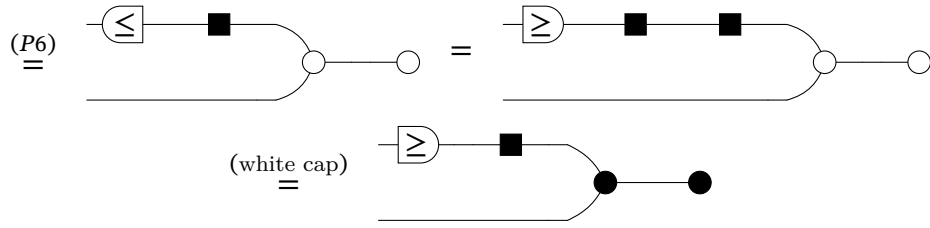
Lemma 1.2 (Cap form). *In $\mathbf{GPA}_{\perp}$,*

$$\begin{aligned}
(C1) \quad & \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} = \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} = \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} \\
(C2) \quad & \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} = \text{---} \boxed{\geq} \text{---} \text{---} \blacksquare \text{---}
\end{aligned}$$

The first chain reads $x + u \geq v$ iff $v - u \leq x$: an inequality may be applied before or after a sum indifferently. The second reads $s + u \geq 0$ iff $s \geq -u$, and it is the form we shall use: the diagram $(\text{---} \boxed{\geq} \text{---} \oplus \text{id}); \varepsilon$, whose denotation is $\{(\ell, r) \mid \ell \geq r\}$, we call the *cap form* of $\ell \geq r$, so that a row $s + u \geq 0$ of a system is the cap form of the bound $s \geq -u$.

Proof. Bending the second input of the white sum to an output is a bijection on hom-sets, so the first equation may be compared in bent form; the third follows by commutativity of the sum. Closing the bent leg with a white cozero then gives the second equation.

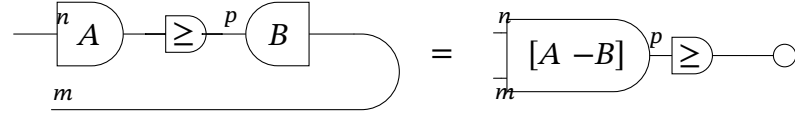
$$\begin{aligned}
(C1) \quad & \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} = \text{---} \boxed{\geq} \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} = \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} \\
& = \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} \boxed{\geq} \text{---} = \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} \\
(C2) \quad & \text{---} \bigcirc \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} \stackrel{(C1)}{=} \text{---} \boxed{\geq} \text{---} \text{---} \bigcirc \text{---} \text{---} \bigcirc \text{---}
\end{aligned}$$



□

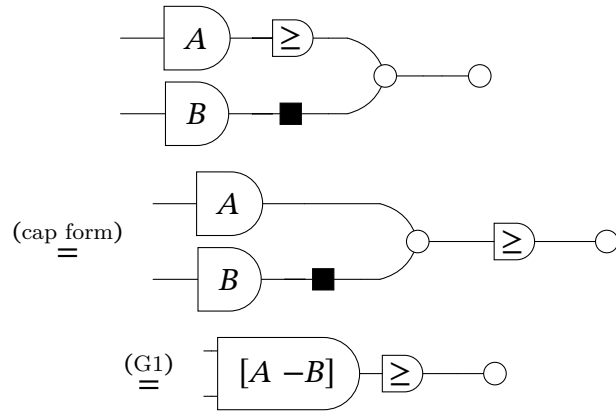
The same bending turns a two-sided system $Ax \geq By$ into a homogeneous one. This is the presentation in which elimination takes place, since it has a single block of rows and no distinguished side.

Lemma 1.3 (Cone form). *Let $A : n \rightarrow p$ and $B : m \rightarrow p$ be matrices. Bending the m outputs of $A; -[\geq]; B^{\text{op}}$ into inputs yields the cone $[A \ -B]; -[\geq]^{\oplus p}; \text{coz}^{\oplus p}$, and unbending recovers the original diagram.*



That is, $\exists v. Ax \geq v, v = By$ iff $Ax - By \geq 0$. Rescaling the rows of a cone by positive constants leaves it unchanged, since $Dz \geq 0$ iff $z \geq 0$ for diagonal $D > 0$; this is (P5) again, applied row by row.

Proof. Bending the comatrix exposes a white sum with an antipode on one leg, and the cap form pushes the inequality past it.



□

The third move, the pairwise merge, is (spider). It is the only one of the three that is not local: on its left-hand side a black node forces a single intermediate value below every x_i and above every y_j , and on its right-hand side that value is gone and only the relations $x_i \geq y_j$ survive. The axiom is therefore the elimination of an intermediate variable in the form discussed in the introduction, and the number of rows grows from $a + b$ to ab exactly as it does in the classical algorithm.

1.4 FOURIER–MOTZKIN ELIMINATION

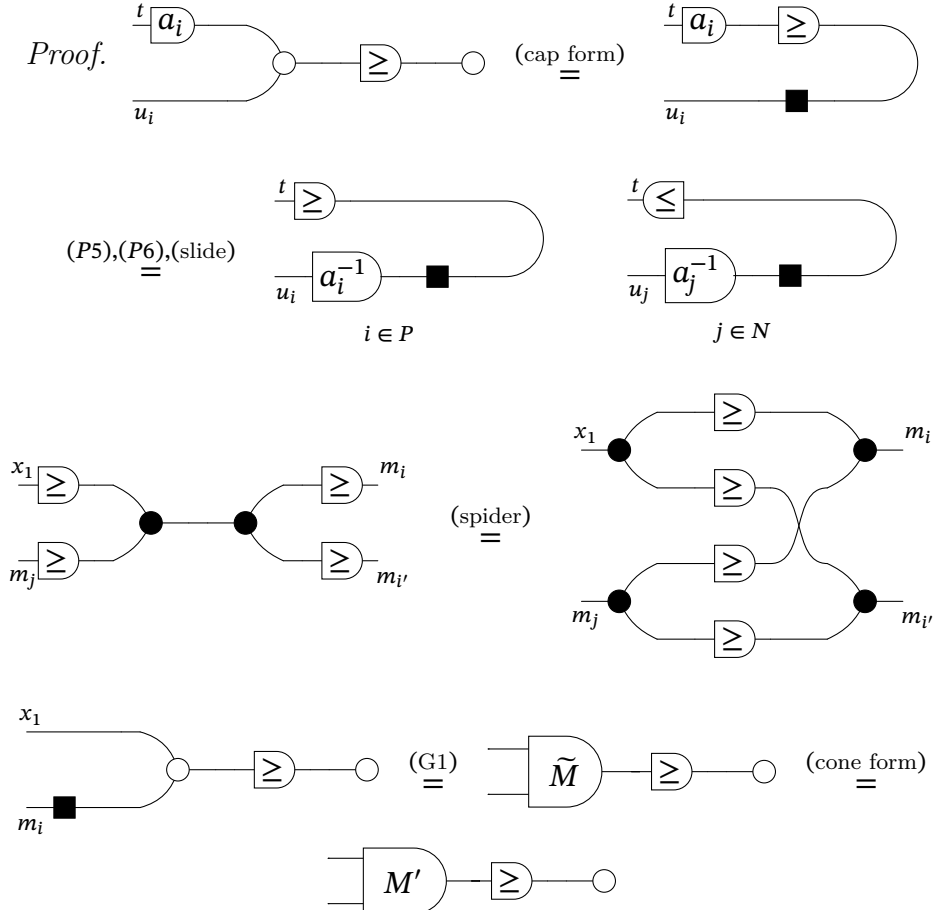
With the three moves available we can eliminate a single variable. Let $M \in \mathbb{R}^{p \times q}$ have first column a , and split the row indices into $P = \{i \mid a_i > 0\}$, $N = \{j \mid a_j < 0\}$ and $Z = \{i \mid a_i = 0\}$.

Lemma 1.4 (One elimination step). *Let M' be the matrix with rows M_i for $i \in P \cup Z$ and $|a_j|M_i + a_i M_j$ for $(i, j) \in P \times N$, so that $p' = |P| + |Z| + |P||N|$. Then*

$$\begin{array}{c} 1 \\ \geq \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} p \\ \geq \end{array} \text{---} \bigcirc = \begin{array}{c} 1 \\ \geq \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} p' \\ \geq \end{array} \text{---} \bigcirc$$

$q-1$ $q-1$

Write t for the wire leaving the generator, x_r for the remaining inputs, $u_i = (M_r x_r)_i$ and $m_i = -u_i/a_i$. Row i of the system reads $a_i t + u_i \geq 0$; the rows in Z do not mention t and are carried along unchanged.



The first line brings a row to cap form and divides by its coefficient, so that it becomes $t \geq m_i$ for $i \in P$ and $t \leq m_j$ for $j \in N$. The second merges the bounds by (spider), the m_j legs having been slid across their caps so that the node carries an inequality on every leg. The third relinearises the resulting pairs and rescales them by the positive constants a_i and $a_i|a_j|$. When $P = \emptyset$ every bound is discarded by (P3), which is the statement that t

may be chosen arbitrarily small; when $N = \emptyset$ nothing is bent and the node is just copied by (P1). $\square$

Iterating the step eliminates the variables bound by the outer generators one at a time, which is the diagrammatic FME.

Proposition 1.1 (Fourier-Motzkin Elimination). *The following diagrammatic procedure is equivalent to perform the FME algorithm*

$$\text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} \boxed{A} \text{---} \boxed{B} \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} = \text{---} \boxed{A'} \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} \boxed{B'} \text{---}$$

A pair (x, y) is related by $\text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}; g; \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}$ precisely when there are v, w with $x \geq v$, $(v, w) \in \llbracket g \rrbracket$ and $w \geq y$, so the two bound variables v, w are what the elimination has to remove.

$$\begin{aligned} \text{Proof.} \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} \boxed{g} \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^m \\ (G2) \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^n \boxed{E} \text{---}^k \boxed{F} \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^m \\ (bend) \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^n \text{---} \begin{array}{|c|} \hline [E - F]^k \\ \hline \end{array} \text{---} \begin{array}{|c|} \hline \leq \\ \hline \end{array} \text{---}^m \\ (antipode) \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^n \text{---} \boxed{M_1} \text{---}^{2k} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} \text{---}^m \\ (FME) \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^n \text{---} \boxed{M_2} \text{---}^p \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---} \text{---}^m \\ (unbend) \quad & \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^n \boxed{A'} \text{---} \begin{array}{|c|} \hline \geq \\ \hline \end{array} \text{---}^p \boxed{B'} \text{---}^m \end{aligned}$$

The cospan form (G2) presents the middle condition as an equation, which the cone form (Lemma 1.3) and the antipode turn into a single block M_1 of $2k$ rows; Lemma 1.4, applied $n + m$ times, eliminates the bound variables and leaves M_2 ; unbending the last m inputs returns the claimed shape. The rows coming from the middle equations are never eliminated and simply survive in the final cone. $\square$

With FME available as a diagrammatic operation, we can now prove the normal form theorem for $\mathbf{GPA}_+$. We proceed by structural induction, handling generators, composition, and tensor product in turn, using FME steps to push all inequality generators into a canonical block form.

Theorem 1.1 (First Normal Form). *Every diagram in $\mathbf{GPA}_+$ admits an isomorphic representation, modulo the axioms, of the following form:*

$$\text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---}$$

Proof. We proceed by structural induction. The two base cases are as follows. First, the $\geq$ generator:

$$\boxed{\geq} = \boxed{} \boxed{\geq} \boxed{} = \text{---} \boxed{I} \text{---} \boxed{\geq} \text{---} \boxed{I} \text{---}$$

Second, for any diagram c in $\mathbf{GLA}$:

$$\begin{aligned} \text{---} \boxed{A} \text{---} &= \text{---} \boxed{A} \text{---} \bullet \begin{array}{c} \boxed{\geq} \\ \boxed{\leq} \end{array} \bullet \text{---} \\ &= \bullet \begin{array}{c} \boxed{A} \boxed{\geq} \\ \boxed{A} \boxed{\leq} \end{array} \bullet \quad (\text{antipode}) \quad \bullet \begin{array}{c} \boxed{A} \boxed{\geq} \\ \boxed{-A} \boxed{\geq} \end{array} \bullet \\ &\stackrel{(\text{matrix def.})}{=} \text{---} \boxed{\bar{A}} \text{---} \boxed{\geq} \text{---} \boxed{\bar{B}} \text{---} \end{aligned}$$

For the inductive step, we consider composition of two diagrams $c; d$ in $\mathbf{GPA}_+$. By the inductive hypothesis, both c and d are already in normal form:

$$\begin{aligned} &\text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---} \quad ; \quad \text{---} \boxed{C} \text{---} \boxed{\geq} \text{---} \boxed{D} \text{---} \\ &= \text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B \quad C} \text{---} \boxed{\geq} \text{---} \boxed{D} \text{---} \\ &\stackrel{(FME)}{=} \text{---} \boxed{\hat{A}} \text{---} \boxed{\geq} \text{---} \boxed{\hat{D}} \text{---} \end{aligned}$$

And for the tensor product:

$$\begin{array}{c} \text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---} \\ \text{---} \boxed{C} \text{---} \boxed{\geq} \text{---} \boxed{D} \text{---} \end{array} = \text{---} \boxed{\bar{A}} \text{---} \boxed{\geq} \text{---} \boxed{\bar{B}} \text{---}$$

□

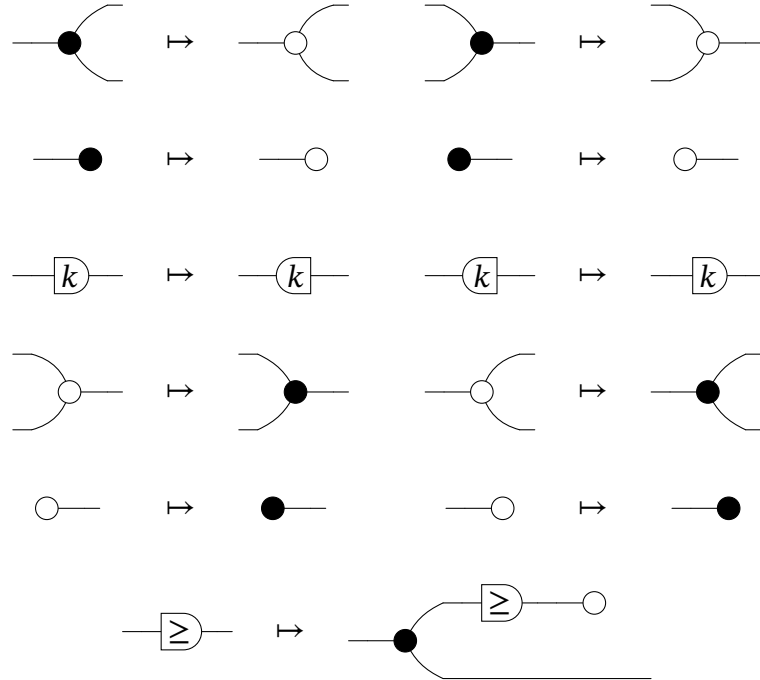
Working with a normal form defined up to isomorphism is in principle weaker than requiring syntactic equality of terms. This choice is motivated by the compact-closed structure, which provides a canonical bijection $\text{Hom}(A, B) \cong \text{Hom}(1, A^* \otimes B)$, allowing input wires to be treated symmetrically with output wires but only up to this isomorphism, not up to equality. As shown in [BDS21], the same completeness argument can be carried out using strict diagrammatic equality; however, the derivations become significantly more involved. Since every result established for the isomorphic normal form carries over to the

strict one, the only difference being the need to bend wires at each step, we adopt the more convenient convention throughout.

This idea was already encountered in the treatment of Graphical Quadratic Algebra [Ste+24], but it is worth pausing to appreciate its significance. The compact-closed structure does not merely allow a more convenient notation; it enables diagrammatic reasoning to serve as a rigorous replacement for coordinate-based algebraic manipulation while remaining extremely intuitive: bending a wire to the left or to the right does exactly what one would expect. Even more, a statement that would classically require several lines of matrix algebra, tracking indices, transposing, and substituting, can instead be verified by a sequence of local wire manipulations, each justified by a single axiom.

This first normal form expresses every diagram as a system of inequalities applied to a linear map. A dual normal form can be obtained by applying the polar operator, which we now define. The polar operator plays the role of duality in the polyhedral setting, exchanging the H -representation (intersection of halfspaces) with the V -representation (conic hull of generators), in direct analogy with the Farkas lemma [BDZ21].

Definition 1.6 (Polar Operator). *The polar operator $-^\circ$ is the involution on generators defined by:*



Its action on diagrams is defined functorially by $(a \circ b)^\circ = a^\circ \circ b^\circ$.

Proposition 1.2 (Second Normal Form). *Every diagram $s \in \mathbf{GPA}_+$ can be written as:*

$$\left(\text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---} \right)^\circ = \text{---} \boxed{A^T} \text{---} \bullet \begin{cases} \boxed{\geq} \text{---} \circ \\ \boxed{B^T} \text{---} \end{cases}$$

Proof. Apply the polar operator to the first normal form of every diagram $c \in \mathbf{GPA}_{\perp}$. Since the polar operator is an involution and acts functorially, the result is again a valid diagram in $\mathbf{GPA}_{\perp}$ in the claimed form. $\square$

The second normal form expresses every diagram as a conic combination of column vectors, which is precisely the V -representation of a polyhedral cone. This is the representation that will drive the completeness proof: two diagrams that are semantically equal must have the same conic hull, and the axioms of $\mathbf{GPA}_{\perp}$ are exactly sufficient to witness this equality diagrammatically.

1.5 PRESENTATION RESULTS FOR $\mathbf{PCREL}$

We now develop the presentation theory for the homogeneous case, proceeding to establish the three properties that together yield the presentation of $\mathbf{PCRel}$.

Lemma 1.5 (Soundness). *If $s = t$ in $\text{FreeP}_{\mathbf{GPA}_{\perp}}$, then $[s] = [t]$ in $\mathbf{PCRel}$.*

Proof. All axioms except those involving the newly introduced $\geq$ generator are already present in $\mathbf{GAA}$ and therefore verified. It suffices to check that the new axioms preserve the semantics:

(P1)

$$\begin{aligned} \left[\text{---} \boxed{\geq} \text{---} \bullet \begin{cases} \text{---} \\ \text{---} \end{cases} \right] &= \left\{ (x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}) \mid x \geq y_1 \wedge x \geq y_2 \wedge y_1 = y_2 \right\} \\ &\subseteq \left\{ (x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}) \mid x \geq y_1 \wedge x \geq y_2 \right\} = \left[\text{---} \bullet \begin{cases} \boxed{\geq} \text{---} \\ \boxed{\geq} \text{---} \end{cases} \right] \end{aligned}$$

(P2)

$$\begin{aligned} \left[\begin{pmatrix} \text{---} \\ \text{---} \end{pmatrix} \text{---} \boxed{\geq} \text{---} \right] &= \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \right) \mid x_1 + x_2 \geq y \right\} = \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \right) \mid \exists c, d. x_1 + x_2 \geq c + d \wedge c + d = y \right\} \\ &= \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \right) \mid \exists \hat{c}, \hat{d}. x_1 \geq \hat{c} \wedge x_2 \geq \hat{d} \wedge \hat{c} + \hat{d} = y \right\} = \left[\begin{cases} \boxed{\geq} \text{---} \\ \boxed{\geq} \text{---} \end{cases} \text{---} \begin{pmatrix} \text{---} \\ \text{---} \end{pmatrix} \right] \end{aligned}$$

(P3)

$$\left[\text{---} \boxed{\geq} \text{---} \bullet \right] = \{(x, \bullet) \mid \exists c. x \geq c\} \underset{\text{tautology}}{=} \{(x, \bullet) \mid x \in \mathbb{R}\} = \left[\text{---} \bullet \right]$$

(P4)

$$\left[\bigcirc \text{---} \right] = \{(x, \bullet) \mid x = 0\} \subseteq \{(x, \bullet) \mid x \geq 0\} = \left[\bigcirc \text{---} \boxed{\geq} \text{---} \right]$$

(P5), (P6)

For $k > 0$:

$$\begin{aligned} \left[\text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} \right] &= \{(x, y) \mid kx \geq y \wedge k > 0\} = \{(x, y) \mid x \geq \frac{y}{k} \wedge k > 0\} \\ &= \{(x, y) \mid \exists c. x \geq c \wedge kc = y \wedge k > 0\} = \left[\text{---} \boxed{\geq} \text{---} \boxed{k} \text{---} \right] \end{aligned}$$

For $k < 0$: the same computation, dividing by the negative k flips the inequality, so

$$\left[\text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} \right] = \left[\text{---} \boxed{\leq} \text{---} \boxed{k} \text{---} \right]$$

(ANTISYM)

$$\begin{aligned} \left[\text{---} \bullet \begin{array}{c} \boxed{\geq} \\ \boxed{\leq} \end{array} \bullet \text{---} \right] &= \{(x, y) \mid x \geq y \wedge x \leq y\} \\ &= \{(x, y) \mid x = y, x, y \in \mathbb{R}\} \subseteq \{(x, y) \mid x = y, x, y \in \mathbb{R}\} = \left[\text{---} \right] \end{aligned}$$

(SPIDER)

$$\left[\begin{array}{c} \boxed{\geq} \\ \boxed{\geq} \end{array} \bullet \text{---} \bullet \begin{array}{c} \boxed{\geq} \\ \boxed{\geq} \end{array} \right] = \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \mid \exists z. x_1 \geq z \wedge x_2 \geq z \wedge z \geq y_1 \wedge z \geq y_2 \right\}$$

$$= \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \mid x_1 \geq y_1 \wedge x_1 \geq y_2 \wedge x_2 \geq y_1 \wedge x_2 \geq y_2 \right\} = \left[\begin{array}{ccc} & \begin{array}{c} \geq \\ \geq \end{array} & \\ \bullet & & \bullet \\ & \begin{array}{c} \geq \\ \geq \end{array} & \\ \bullet & & \bullet \end{array} \right]$$

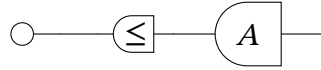
(DIRECTION)

$$\left[\begin{array}{c} \leq \\ \geq \end{array} \right] = \{(x, y) \mid \exists z. z \geq y \wedge z \geq x\} = \{(x, y) \mid x, y \in \mathbb{R}\} = \left[\begin{array}{cc} \bullet & \bullet \end{array} \right]$$

□

Lemma 1.6 (Expressivity). *For every $f : m \rightarrow n$ in $\mathbf{PCRel}$ there exists a Σ -term $s : m \rightarrow n$ in $\text{Hom}_{\text{FreeP}_{\mathbf{GPA}}}(m, n)$ such that $[s] = f$.*

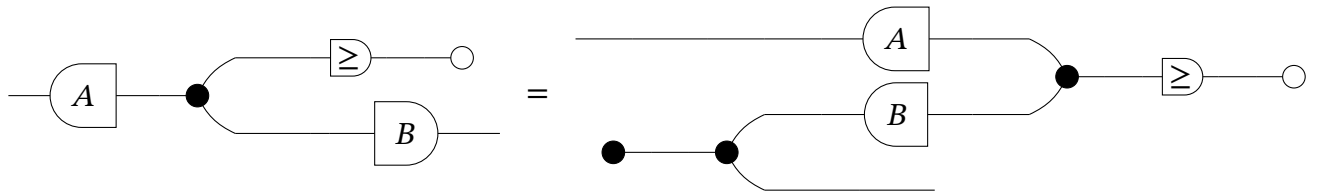
Proof. By Definition 1.1, every polyhedral cone has the form $\{x \mid Ax \geq 0, x \in \mathbb{R}^n\}$. This is precisely the interpretation of the diagram:



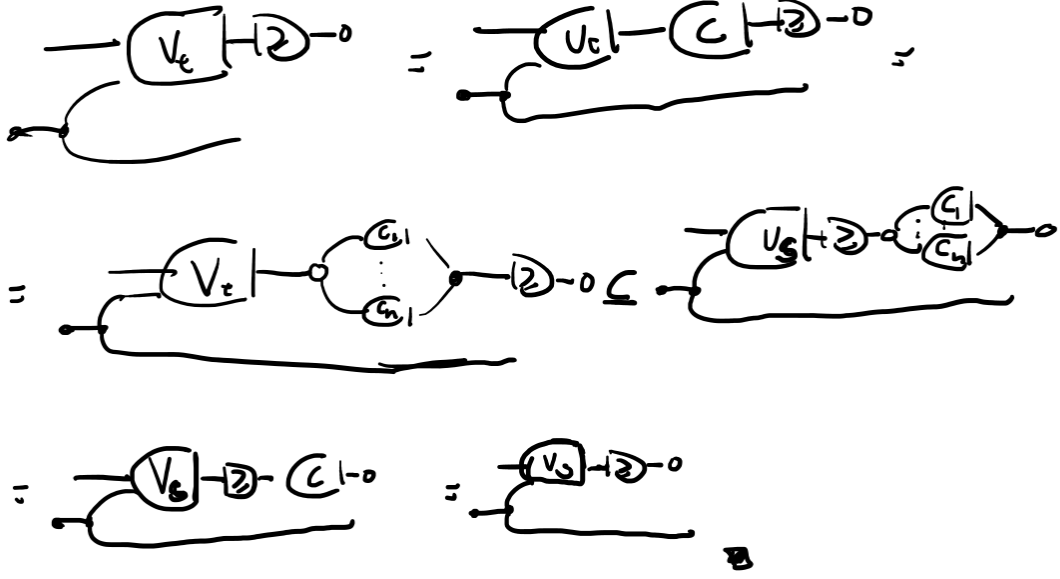
□

Lemma 1.7 (Completeness). *If $[s] = [t]$ in $\mathbf{PCRel}$, then $s = t$ in $\text{FreeP}_{\mathbf{GPA}}$.*

Proof. By the second normal form, we write s and t in V -representation form:



The semantics of such a diagram is the conic hull of the column vectors of V . Since $[s] = [t]$ implies in particular $[s] \subseteq [t]$, the conic combinations of V_t are a subset of those of V_s . Hence there exists a non-negative matrix C such that $CV_t = V_s$. The diagrammatic witness is:



The inclusion $[s] \subseteq [t]$ follows by the same argument, yielding equality. $\square$

Theorem 1.2. *PCRel is presented by $\text{FreeP}_{\text{GPA}}^{\perp}$.*

Proof. Soundness, expressivity, and completeness were established by the preceding three lemmas. Together they imply that the canonical PROP morphism is an isomorphism. $\square$

This settles the homogeneous case: every polyhedral cone is diagrammatically expressible, and two diagrams denote the same cone exactly when the axioms can prove them equal. Cones are, however, a narrow class, they always contain the origin, so they cannot represent an arbitrary feasible region or, later, an arbitrary linear program. The next step lifts this result from cones to general polyhedra, replacing the homogeneous inequalities $Ax \geq 0$ with the affine $Ax \geq b$, and the machinery developed above will carry over almost unchanged.

1.6 SEMANTICS: THE PROP **POLYREL** OF POLYHEDRA

Now we start the presentation of the broader class of polyhedral relations by recalling the definition of a polyhedron.

Definition 1.7. *A set $P \subseteq \mathbb{R}^n$ is called a polyhedron if there exist $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ such that*

$$P = \{x \in \mathbb{R}^n \mid Ax \geq b\}.$$

Note that a polyhedral cone is a special case of a polyhedron with $b = 0$. The distinction is algebraically significant: cones are closed under non-negative scaling and admit a purely homogeneous representation, while general polyhedra may be translated.

This distinction is reflected in the two graphical algebras we now define, where $\mathbf{GPA}_+$ handles the homogeneous case and $\mathbf{GPA}$ extends it with an affine generator.

In that light, one can define the category of polyhedral relations as:

Definition 1.8 (The prop $\mathbf{PolyRel}$ of polyhedra). *Let k be an ordered field. The prop $\mathbf{PolyRel}_k$ is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n .*
- *A morphism $n \rightarrow m$ is a polyhedron $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$, $A \in k^{p \times (n+m)}$, and $b \in k^p$ such that*

$$R = \left\{ (x, y) \in k^n \times k^m \left| A \begin{pmatrix} x \\ y \end{pmatrix} + b \geq 0 \right. \right\}.$$

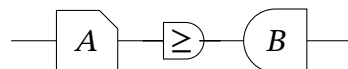
- *Composition is relational composition.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the Cartesian product.*

Notice how, the same way that $\mathbf{PCRel}$ offered us a more relational way to think about Polyhedral Cones, so does $\mathbf{PolyRel}$. Indeed, one may think of polyhedrals as closed spaces in some geometry that respects some linear constraints. We argue, however, that this definition is overly convoluted, and a more natural way to denote such objects is to think in terms of relations R , such that $xRy \iff Ax + b \geq By + c$ for some matrices A, B and vectors b, c . Once again, we move from a set-theoretic heavy description to a more logical one.

1.7 SYNTAX: THE ALGEBRA $\mathbf{GPA}$

We now extend the theory from polyhedral cones to general polyhedra by reintroducing the affine generator $\vdash$. Geometrically, this corresponds to allowing translations: a polyhedron is a translated polyhedral cone, and the generator $\vdash$ provides the syntactic means to encode the right-hand side vector b in $Ax \geq b$. The normal form, soundness, and expressivity arguments all extend smoothly from the homogeneous case.

Theorem 1.3 (Normal Form). *For every diagram c in $\mathbf{GPA}$, there exists a diagram d such that $c = d$ modulo the axioms and d is of the form:*



Proof. The derivation proceeds directly. Let c be a diagram of **GPA**. Then we can collect all $\vdash$ on the left such that c' is a diagram of **GPA** $_{\vdash}$

$$\boxed{c} = \boxed{\vdash} \boxed{c'}$$

From Theorem 1.1 we know that c' admits a normal form in terms of matrices A and B . Breaking down the matrix A in columns we get:

$$\begin{aligned} & \boxed{\vdash} \boxed{A} \boxed{\geq} \boxed{B} = \\ & = \boxed{\vdash} \boxed{A_1} \boxed{A_2} \boxed{\geq} \boxed{B} \end{aligned}$$

thus the thesis. $\square$

1.8 PRESENTATION RESULTS FOR **POLYREL**

Lemma 1.8 (Soundness). *If $s = t$ in $\text{FreeP}_{\mathbf{GPA}}$, then $[s] = [t]$ in **PolyRel**.*

Proof. This follows immediately from the soundness of **PCRel**. Since the generator $\vdash$ introduces no new axioms beyond those already verified for **PCRel** and **GAA**, and both of those theories are sound in **PolyRel**, soundness carries over directly. $\square$

Lemma 1.9 (Expressivity). *For every $f : m \rightarrow n$ in **PolyRel** there exists a Σ -term $s : m \rightarrow n$ in $\text{Hom}_{\text{FreeP}_{\mathbf{GPA}}}(m, n)$ such that $[s] = f$.*

Proof. By Definition 1.7, every polyhedron has the form $\{x \mid Ax \geq b\}$. Appealing to this representation directly, the polyhedron is expressed as the interpretation of the following diagram, where the generator $\vdash$ encodes the right-hand side vector b :

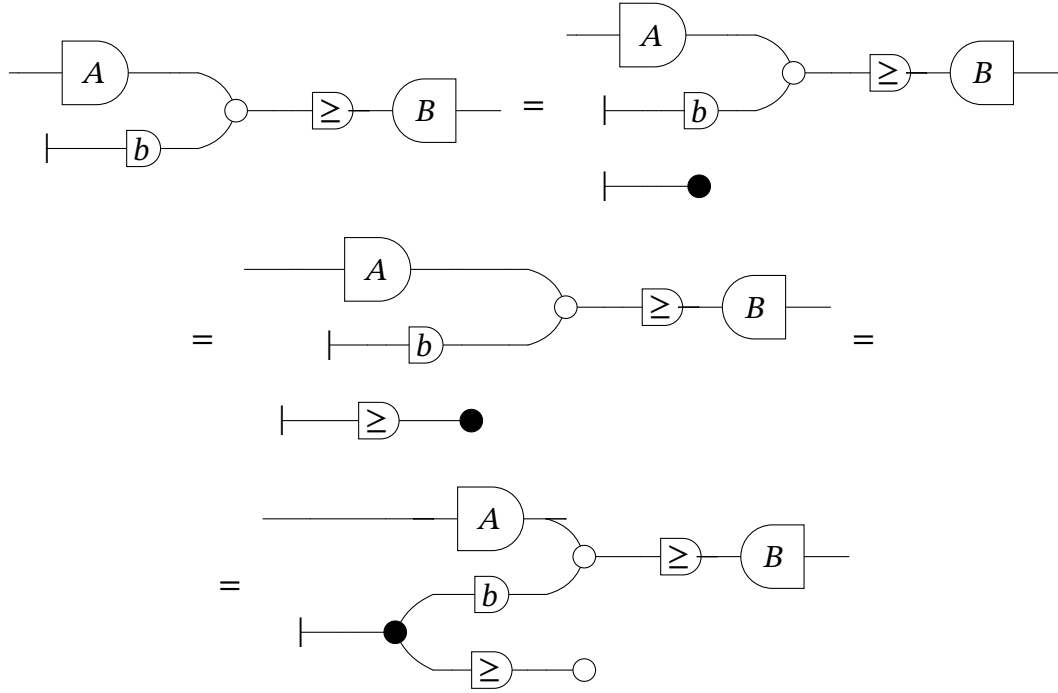
$$\boxed{\vdash} \boxed{b} \boxed{\leq} \boxed{A} \boxed{\vdash} = \{ (b, x) \mid Ax \geq b, x \in \mathbb{R}^n \}$$

$\square$

Lemma 1.10 (Completeness). *If $[s] = [t]$ in **PolyRel**, then $s = t$ in $\text{FreeP}_{\mathbf{GPA}}$.*

Proof. Every polyhedron P can be homogenised by introducing a slack variable, reducing the affine case to the conic one $P^H = \{(x, y) \in k^{n+1} | Ax + by \geq 0, y \geq 0\}$ which admits a normal form (Theorem 1.1). Under this homogenisation, for each two diagrams $s, t \in \mathbf{GPA}$ with $[s] = [t]$ in $\mathbf{PolyRel}$, there are $s', t' \in \mathbf{GPA}_\perp$ such that $[s'] = [s]^H, [t'] = [t]^H$. Since two polyhedrals P, Q are equal iff $P^H = Q^H$, then s and t have equal semantics in $\mathbf{PCRel}$. Completeness of $\mathbf{GPA}_\perp$ then gives equality of the homogenised diagrams, and de-homogenising recovers equality of s and t in $\mathbf{FreeP}_{\mathbf{GPA}}$.

Diagrammatically this goes as follows:



Whereas the interpretation of the last diagram is $\{(x, y, z) \in k^m | Ax + by \geq Bz, y \geq 0\}$, which is the same as P^H . Therefore, the semantics of s' is that of the homogenisation of $[s]$ and follows the above argument. $\square$

Theorem 1.4. $\mathbf{PolyRel}$ is presented by $\mathbf{FreeP}_{\mathbf{GPA}}$.

Proof. By the preceding three lemmas. $\square$

This is the result the rest of the thesis leans on: every polyhedral relation, not just the conic ones from earlier, has a diagrammatic representative, and semantic equality between polyhedra can always be certified by a finite rewriting sequence in $\mathbf{GPA}$. It is this presentation, together with its normal form, that Chapter ?? builds on when it adds a single new generator for the objective function.

Remark 1.1 (Invariance under non-negative matrices). *Whereas the linearity axiom shows the linear generator to be exactly invariant under column-stochastic reweightings (Remark ??), the order generator ge enjoys a related but weaker property with respect to*

non-negative matrices. Let $A \in \mathbb{R}^{n \times m}$ with $A_{ij} \geq 0$ for all i, j . If $x, y \in \mathbb{R}^m$ satisfy $x \geq y$ componentwise, then $Ax \geq Ay$ componentwise: for each row i ,

$$(Ax)_i - (Ay)_i = \sum_j A_{ij}(x_j - y_j) \geq 0,$$

since every term is a product of a non-negative entry and a non-negative difference. Diagrammatically, this is the inclusion

$$\text{matrix}[A] ; \text{ge}^{\otimes n} \subseteq \text{ge}^{\otimes m} ; \text{matrix}[A]^{\otimes 2},$$

verified directly in the semantics of **PolyRel**, exactly as the axioms of **GPA** are themselves verified. Unlike Remark ??, this is only an inclusion and not an equality: a non-negative matrix need not reflect the order, only preserve it, so $Ax \geq Ay$ can hold even when $x \geq y$ fails (take $A = 0$). In other words, a non-negative matrix always goes through the order generator, certifying that any inequality derived in **GPA** remains valid after left multiplication by A .